PERSONAL EXPENSE ANALYTICS

Python-Based Data Analysis Project

Internship Project Report

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Project Report

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1. Abstract

Personal Expense Analytics is a Python-based data analysis project designed to analyze personal spending patterns. The project uses a CSV dataset containing expense details such as date, category, description, amount, and payment mode.

Python libraries such as Pandas and Matplotlib are used to process the dataset, calculate expense summaries, and generate graphical visualizations. The project provides insights into total expenses, average expenses, category-wise spending, monthly expenses, and payment methods.

The main purpose of this project is to help users understand their spending habits through data analysis and graphical representation.

2. Introduction

Managing personal expenses is important for understanding financial habits and controlling unnecessary spending. Many people record their daily expenses but may find it difficult to analyze their spending patterns manually.

Personal Expense Analytics provides a simple solution by using Python to process and analyze expense records. It converts raw expense data into meaningful information through calculations and visualizations.

The project helps users understand how much they spend, which categories consume more money, and how their expenses change over different months.

3. Problem Statement

Managing personal expenses manually can be time-consuming and may not provide clear information about spending patterns. Without proper analysis, users may find it difficult to identify their highest spending categories, compare monthly expenses, and understand their preferred payment methods.

Therefore, this project aims to develop a Python-based system that analyzes personal expense records and presents useful financial insights through statistical summaries and graphical visualizations.

4. Objectives

- To analyze personal spending habits.
- To calculate total and average expenses.
- To identify category-wise spending patterns.
- To analyze monthly expense variations.
- To examine different payment methods.
- To generate graphical representations of expenses.
- To prepare a cleaned dataset for further analysis.

5. Scope of the Project

The project focuses on analyzing personal expense records using Python. It includes:

- Reading expense data from CSV files.
- Processing and organizing the dataset.
- Calculating total and average expenses.
- Performing category-wise analysis.
- Comparing monthly expenses.
- Analyzing payment modes.
- Generating charts and output files.

The project is designed for basic personal financial analysis and can be extended with additional features in the future.

6. Technologies Used

Technology	Purpose
Python	Programming language
Pandas	Data processing and analysis
Matplotlib	Data visualization
CSV	Dataset storage
VS Code	Development environment

7. Dataset Description

Figure 1 shows the sample expense dataset used by the project. The dataset contains transaction date, category, description, amount, and payment mode fields.

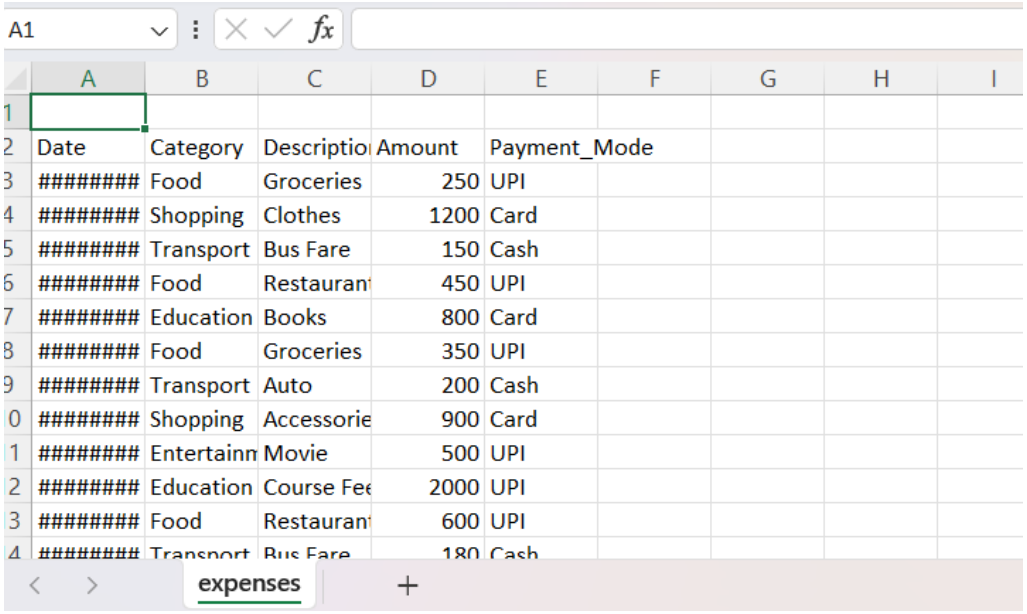


Figure 1: Sample Expense Dataset

Attribute	Description
Date	Date of the transaction
Category	Type of expense
Description	Details of the expense
Amount	Money spent
Payment Mode	Method of payment

Examples of expense categories include Education, Entertainment, Food, Shopping, and Transport. Payment methods include UPI, Cash, and Card.

8. System Requirements

Hardware Requirements

- Processor: Intel or AMD processor
- RAM: Minimum 4 GB
- Storage: Minimum 500 MB free space
- Operating System: Windows 10 or above

Software Requirements

- Python 3
- Visual Studio Code
- Pandas library
- Matplotlib library
- CSV dataset

9. Methodology

The project follows a simple data analysis process.

Step 1: Dataset Collection

Expense information is stored in a CSV file.

Step 2: Data Loading

The Pandas library reads the dataset into a DataFrame.

Step 3: Data Processing

The records are prepared for analysis, and a cleaned dataset is generated.

Step 4: Data Analysis

Calculations are performed to understand total expenses, average expenses, category-wise spending, monthly expenses, and payment modes.

Step 5: Data Visualization

Matplotlib is used to generate graphs for representing the analyzed information.

Step 6: Output Generation

The generated graphs and cleaned dataset are saved in the outputs folder.

10. Analysis Performed

10.1 Total and Average Expense Analysis

This analysis calculates the total amount spent and the average expense per transaction. It helps users understand their overall spending.

10.2 Category-wise Expense Analysis

This analysis groups expenses according to categories such as Food, Shopping, Education, Entertainment, and Transport. A bar chart is generated to compare spending across different categories.

10.3 Monthly Expense Analysis

Monthly expense analysis compares spending across January, February, March, and April. A line chart is used to represent changes in monthly expenses.

10.4 Payment Mode Analysis

This analysis examines expenses based on payment methods such as UPI, Cash, and Card. A pie chart represents the distribution of payment methods.

11. Results and Findings

The project successfully analyzed the expense dataset and generated useful financial insights. The results obtained are:

Metric	Result
Total Expense	■ 13,530
Average Expense	■ 676.50
Highest Spending Category	Shopping
Highest Spending Month	February
Highest Amount by Payment Mode	Card

Monthly Expense Summary

Month	Amount
January	■ 2,850
February	■ 3,950
March	■ 2,980
April	■ 3,750
Total	■ 13,530

These findings help understand spending distribution and monthly variations.

12. Graphical Visualizations

The project generates three graphical visualizations.

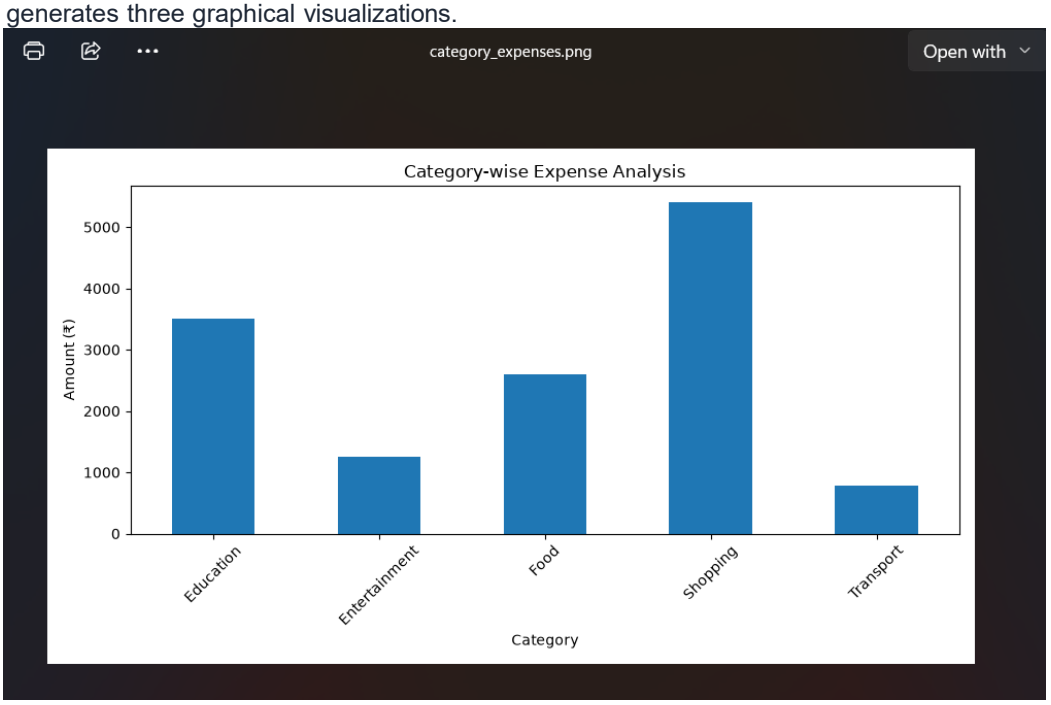


Figure 2: Category-Wise Expense Analysis

This bar chart represents the total amount spent across different expense categories.

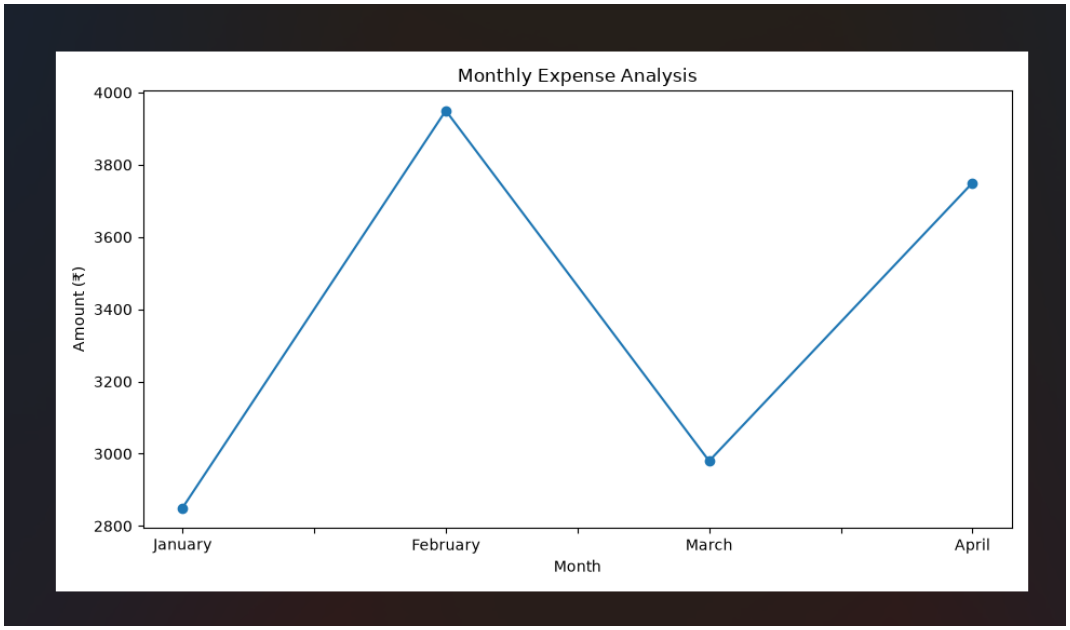


Figure 3: Monthly Expenses Analysis

This line chart displays monthly spending variations.

Payment Mode Distribution

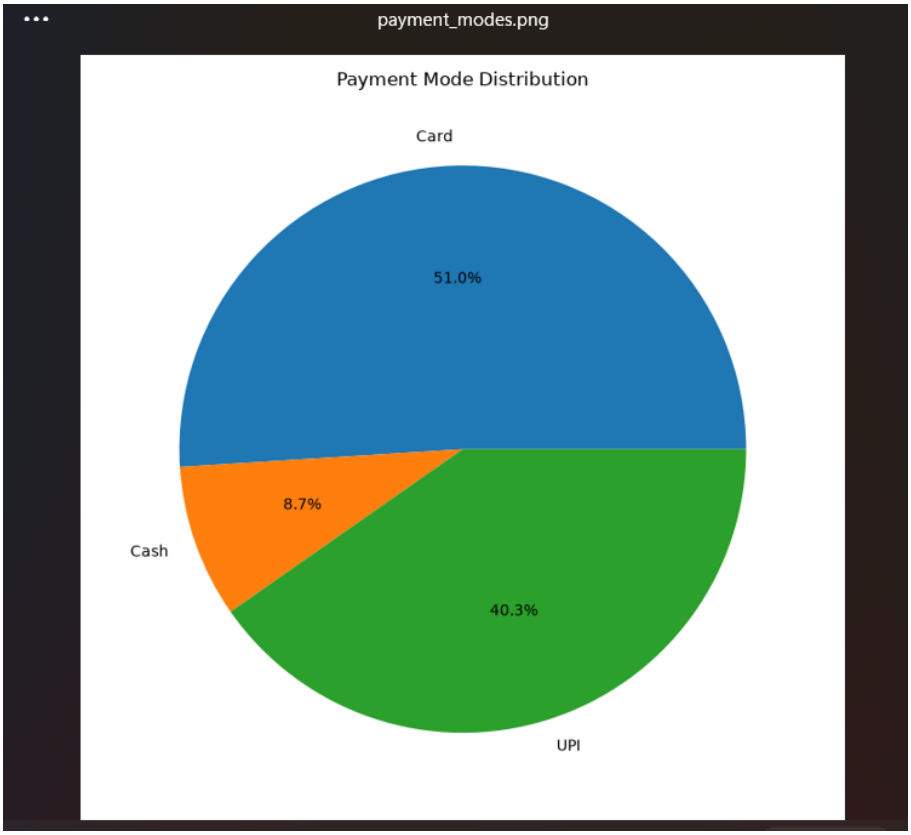


Figure 4: Payment Mode Analysis

This pie chart represents the distribution of expenses across different payment methods.

13. Project Structure

The project is organized into the following directories:

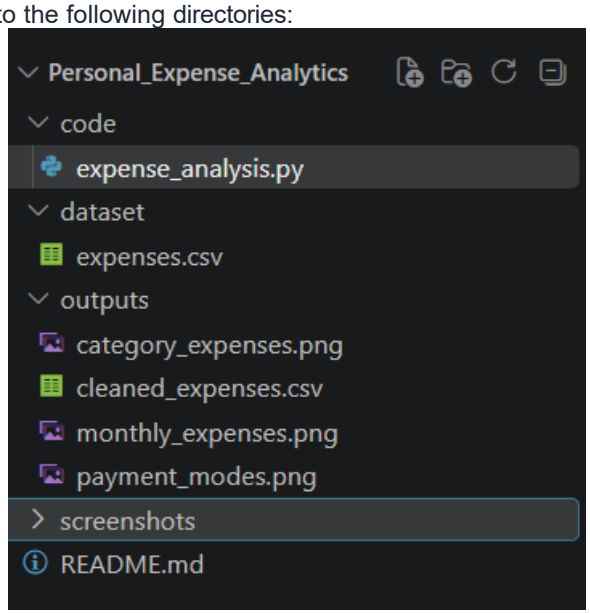


Figure 5: Project Structure

14. Implementation

The project is implemented using Python. Pandas is used to read and process the CSV dataset. Different grouping and calculation operations are performed to analyze expenses. Matplotlib is used to create graphical visualizations. The generated charts and cleaned dataset are stored in the outputs directory.

The following screenshots show the main Python implementation and its analysis sections.

```python
import pandas as pd
import matplotlib.pyplot as plt
import os

# Project paths
base_path = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))

dataset_path = os.path.join(base_path, "dataset", "expenses.csv")
output_path = os.path.join(base_path, "outputs")

os.makedirs(output_path, exist_ok=True)

# Load dataset
df = pd.read_csv(dataset_path)

# Data cleaning
df["Date"] = pd.to_datetime(df["Date"])
df["Amount"] = pd.to_numeric(df["Amount"], errors="coerce")
df.dropna(inplace=True)
df.drop_duplicates(inplace=True)

# Extract month
```

Figure 6A: Python Implementation - Data Loading and Cleaning

```python
# Extract month
df["Month"] = df["Date"].dt.strftime("%B")

# Basic analysis
total_expense = df["Amount"].sum()
average_expense = df["Amount"].mean()

category_expense = df.groupby("Category")["Amount"].sum()
monthly_expense = df.groupby("Month")["Amount"].sum()

print("PERSONAL EXPENSE ANALYTICS")
print("--------------------------")

print("Total Expense: ₹", total_expense)
print("Average Expense: ₹", round(average_expense, 2))

print("\nCategory-wise Expenses:")
print(category_expense)

print("\nMonthly Expenses:")
print(monthly_expense)
```

Figure 6B: Python Implementation - Basic Expense Analysis

```python
# Category-wise bar chart
plt.figure(figsize=(9, 5))
category_expense.plot(kind="bar")
plt.title("Category-wise Expense Analysis")
plt.xlabel("Category")
plt.ylabel("Amount (₹)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig(os.path.join(output_path, "category_expenses.png"))
plt.close()

# Monthly expense chart
monthly_order = ["January", "February", "March", "April"]
monthly_expense = monthly_expense.reindex(monthly_order, fill_value=0)

plt.figure(figsize=(9, 5))
monthly_expense.plot(kind="line", marker="o")
plt.title("Monthly Expense Analysis")
plt.xlabel("Month")
plt.ylabel("Amount (₹)")
plt.tight_layout()
plt.savefig(os.path.join(output_path, "monthly_expenses.png"))
plt.close()
```

Figure 6C: Python Implementation - Category and Monthly Charts

```python
plt.savefig(os.path.join(output_path, "monthly_expenses.png"))
plt.close()

# Payment mode analysis
payment_expense = df.groupby("Payment_Mode")["Amount"].sum()

plt.figure(figsize=(7, 7))
payment_expense.plot(kind="pie", autopct="%1.1f%%")
plt.title("Payment Mode Distribution")
plt.ylabel("")
plt.tight_layout()
plt.savefig(os.path.join(output_path, "payment_modes.png"))
plt.close()

# Save cleaned dataset
df.to_csv(os.path.join(output_path, "cleaned_expenses.csv"), index=False)

print("\nAnalysis completed successfully!")
print("Graphs saved in outputs folder.")
```

Figure 6D: Python Implementation - Payment Mode Analysis and Output

15. How to Run the Project

Follow these steps to execute the project:

- Install Python on your computer.
- Install Visual Studio Code.
- Open the project folder in VS Code.
- Open the terminal.
- Install the required libraries: `pip install pandas matplotlib`
- Execute the Python file: `python code/expense_analysis.py`
- Check the outputs folder for generated graphs and cleaned data.

16. Testing and Execution

The project was executed successfully using Visual Studio Code. The execution process was tested by running the Python program and checking the generated output files. The terminal displayed the monthly expense summary and the successful completion message. The graphs were saved in the outputs folder.

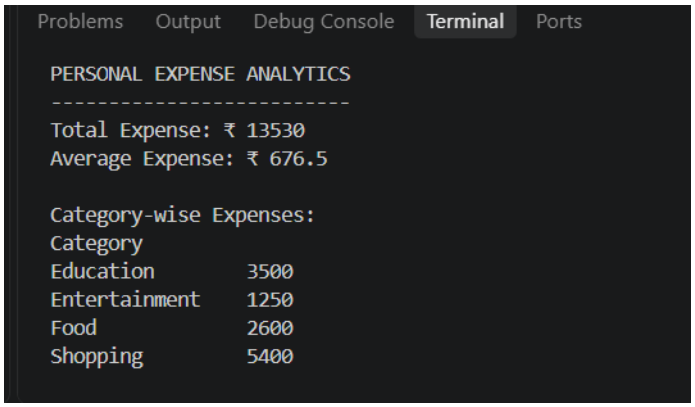


Figure 7A: Project Execution Output - Summary

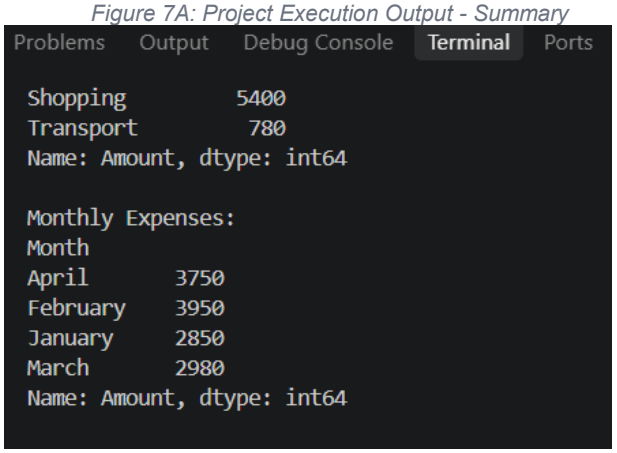


Figure 7B: Project Execution Output - Monthly Results

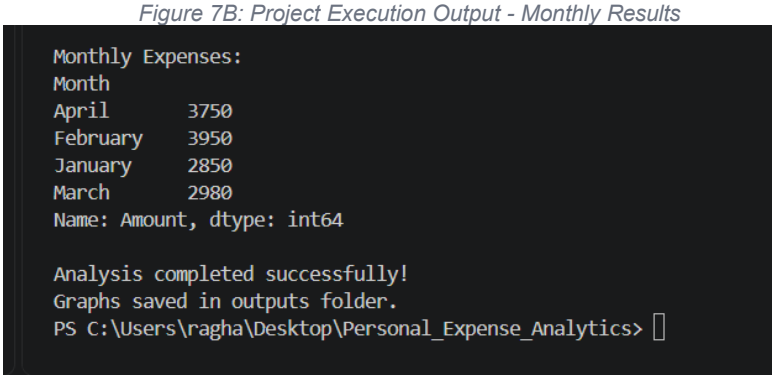


Figure 7C: Project Execution Output - Completion Message

17. Advantages

- Simple and easy-to-understand implementation.
- Reduces manual expense calculations.
- Provides category-wise spending information.
- Helps compare monthly expenses.
- Generates graphical representations.
- Supports CSV-based data analysis.
- Can be extended with additional features.

18. Limitations

- Analysis depends on the available dataset.
- The project uses manually maintained expense records.
- It does not automatically connect to bank accounts.
- It provides basic descriptive analysis.
- It does not include real-time financial tracking.

19. Future Enhancements

The project can be improved by adding:

- Interactive expense dashboards.
- Additional expense categories.
- Date-wise filtering.
- Monthly budget comparison.
- Automatic report generation.
- Expense prediction using machine learning.
- A simple graphical user interface.

20. Conclusion

Personal Expense Analytics successfully demonstrates the use of Python for analyzing personal spending information. The project processes expense records and provides meaningful insights through total expense calculations, category-wise analysis, monthly comparisons, and payment mode analysis.

Using Pandas and Matplotlib, the system transforms raw CSV data into understandable summaries and graphical visualizations. Overall, the project provides a simple and effective approach to understanding personal spending patterns and demonstrates the practical application of Python in data analytics.

End of Project Report
Personal Expense Analytics